

Chiranjit Saha

chiranjitsaha.official680@gmail.com | [LinkedIn](#) | [GitHub](#) | [LeetCode](#) | 8620870598

Backend & AI Systems Engineer — Retrieval-Augmented Generation, Distributed Systems, Multi-Agent Architectures

Education

Indian Institute of Information Technology Guwahati

B.Tech in Electronics and Communications Engineering — CGPA: 8.02

Guwahati, India

Jul 2023 – Aug 2027

Experience

Project Intern

May 2026 – Jul 2026

Oracle India Private Limited

Bengaluru, India

Oracle SaaS Support Assist: Enterprise RAG Support Platform | *Python, Go, OCI, Oracle AI Database 26ai, Wayflow, MCP*

2026

- Led a 4-person intern team building an enterprise Agentic Knowledge platform delivering context-aware ticket-resolution recommendations to 5,000+ support engineers.
- Built a high-performance API gateway in **Go (Gin)** with middleware for authentication, rate limiting, request tracing, and structured logging, sustaining very low routing latency across downstream services.
- Architected **semantically-clustered vector DB clusters with decay-weighted bandit re-ranking** at the partition level—a retrieval pattern outside standard RAG literature—layered with hybrid search, hierarchical retrieval, and a **GraphRAG** module on Oracle's native SQL/PGQ property graphs, reaching **95.4% signal recall**.
- Designed a **Wayflow multi-agent architecture** (retrieval, planning, validation, and response-generation agents) with MCP based tool calling, producing evidence-backed, citation-validated responses and Implemented **AI policy and guardrail controls**—Presidio-based PII scrubbing, response validation, and safety filtering—to enforce secure, compliant AI outputs.

Summer Research Intern

May 2025 – Jul 2025

Indian Institute of Technology Guwahati (Supervisor: Dr. Ayon Borthakur)

Guwahati, India

- **Reversible Vision Transformer (minRev)** | *PyTorch* | [GitHub](#) — Built a memory-efficient Reversible ViT achieving a **50% reduction in GPU memory** vs. a standard ViT baseline while matching classification performance.

Summer Intern

May 2024 – Jul 2024

IIIT Guwahati (Supervisor: Dr. Sudip Biswas)

Guwahati, India

- Built a full-stack IoT management platform (**Django REST Framework**) integrating **NodeMCU, Arduino, Raspberry Pi** for real-time remote monitoring and control of an automated cooking machine.

Projects

Blind Date: Real-Time Anonymous Matchmaking Platform [Collaborative] | *Go, Cassandra, Redis, WebSockets, H3* [Link](#) Aug 2026 – Present

- Implemented H3-based geospatial matchmaking using gridDisk neighbor search and hierarchical roll-up for scalable location-aware matching.
- Built a Go WebSocket engine for anonymous 1:1 chat with per-connection reader/writer goroutines, **RWMutex**-guarded hub, non-blocking fan-out, cursor-based sync, and idempotent reconnects.

VoiceCart: AI Voice Assistant for E-Commerce [Collaborative] | *FastAPI, PostgreSQL, MCP, ReactJS LangGraph* [Link](#) Jun 2025 – July 2025

- Architected a multi-agent conversational commerce platform enabling speech-driven product search, cart management, and order placement through LangGraph orchestration and WebSockets.
- Built independently deployable microservices supporting high-concurrency voice sessions with low-latency communication.

DisasterLM: Decoder-Only Language Model (14.3M Parameters) | *PyTorch, Transformers* [Link](#)

May 2025 – Jun 2025

- Built a 14.3M-parameter decoder-only Transformer from scratch, implementing the complete training pipeline including data preprocessing, BPE tokenization, causal language modeling, cosine learning-rate scheduling, and gradient clipping.
- Pretrained on BookCorpus and fine-tuned on disaster-response datasets, achieving a perplexity of **174** on domain-specific evaluation benchmarks.

Technical Skills

Languages: Python, C++, Golang, Java, SQL, JavaScript

Backend & APIs: FastAPI, Go Gin, Django, Flask, Express JS
WebSockets, REST, Microservices

Databases, Cloud & Infra: OCI, Oracle Autonomous AI DB(26ai), PostgreSQL, MongoDB, ChromaDB, FAISS, Redis, Docker, Podman

AI/ML: RAG, GraphRAG, LangChain, LangGraph, PyTorch, TensorFlow, NLP, Computer Vision

Frontend: ReactJS, Streamlit

Specializations: Distributed Systems, System Design, Multi-Agent Systems, Agent Orchestration

Tools: Git, Linux

Research

Undergraduate Researcher

Jan 2026 – Present

IIIT Guwahati (Supervisor: Dr. Sovan Barma)

Guwahati, India

- **Multi-Agent Debate Framework for Critical Decision-Making** (Jan 2026 – Jul 2026) | *LangGraph, PyTorch* — Designing a debate-based multi-agent system where adversarial LLM agents reason through high-stakes decisions, targeting improved factual consistency and reduced hallucination over single-agent baselines.

Achievements & Leadership

700+ problems solved across competitive programming platforms (LeetCode, CodeChef, Codeforces)

Runner-Up — Inspirathon 2026, IIIT Guwahati

2026

Finalist (200+ teams) — IIT Guwahati Science & Technology Hackathon

2024

PR & Corporate Relations Head, I&E Cell, IIIT Guwahati

Nov 2024 – Dec 2025

Coordinator, Mavericks (ML Society), IIIT Guwahati

Aug 2024 – Dec 2024